# Santiago Morandi

*Curriculum Vitae*


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| Bio | Passionate scientist and chemical engineer with a strong interest in AI and automation. With a strong problem-solving attitude, I enjoy working on AI-driven and first-principles solutions to advance science. I excel in international environments and embrace new challenges with ambition and enthusiasm. |
| Current Location | Institute of Chemical Research of Catalonia (ICIQ), Tarragona, Spain |
| Birth | 29 December 1995, Magenta (Milan, Italy) |
| Nationality | Italian |
| E-Mail | santiagomorandi@yahoo.it |
| Mobile number | +39 3489600390 |


## Experience & Education

**Postdoctoral Researcher** *Institute of Chemical Research of Catalonia*, Tarragona, Spain
**07/2025 – present**
Researcher in the López research group. Member of the EU-funded project SuperVaL (SUstainable Photo-ElectRochemical VALorization of flue gases).

**PhD Researcher** *Institute of Chemical Research of Catalonia*, Tarragona, Spain
**10/2020 – 07/2025**
Doctoral researcher in the group of Prof. Núria López focused on development of multi-scale frameworks for modeling heterogeneous catalysis by means of atomic-scale simulations, Machine Learning (ML), and automation. Developed graph neural networks (GNNs) to accelerate atomistic simulations and algorithms for automated generation of chemical reaction networks. Funded member of NCCR Catalysis project (Switzerland). Author of 5 publications, direct supervisor of 3 M.Sc. students. Juan Bertrán i Rusca 2024 Award from the Spanish Royal Society of Chemistry. Final doctorate grade: Excellent *Cum Laude*.

**Visiting Researcher** *IBM Research Europe*, Zurich, Switzerland
**09/2024 – 12/2024**
Visiting stay at the Artificial Intelligence for Scientific Discovery team lead by Dr. Teodoro Laino. Developed large language models (BART sequence-to-sequence and BERT regression) for material property prediction and text representations for molecules on surfaces.

**Internship** *BASF Antwerpen, Advanced Process Control team*, Antwerp, Belgium
**02/2019 – 09/2019**
Designed and implemented a Nonlinear Model Predictive Control (NMPC) strategy based on a first-principles digital twin of an industrial alkoxylation semi-batch process, with the final aim of reducing the reaction time and increasing plant sustainability.

**Chemical Engineering M.Sc.** *Politecnico di Milano*, Italy
**09/2017 – 12/2019**
Thesis title: "Development of a closed-loop Nonlinear Model Predictive Control simulator for an industrial semi-batch plant". Final grade 110/110 *Cum Laude* (GPA=4.0)

**Chemical Engineering B.Sc.** *Politecnico di Milano*, Italy
**09/2014 – 10/2017**
Thesis title: "Membranes: Industrial Applications".

## Skills

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| Programming | Python, C++, CUDA, Bash, Julia for acceleration of ordinary differential equations |
| DevOps | Git, Github, GitHub Actions, Conda, uv, Docker, Dask |
| AI | Pytorch, Pytorch Geometric, HuggingFace Transformers, Ray Tune for hyperparameter optimization, scikit-learn, uncertainty quantification |
| Visualization | Blender, Inkscape, Matplotlib, Seaborn, Graphviz, Gephi |
| Chemistry tools | VASP, RDKit, ASE, pymatgen |
| Modelling | Industrial process control, dynamic systems, multi-scale reaction kinetics |
| Languages | Italian (native), English (advanced), Spanish (advanced), Catalan (basic), German (scholastic) |

## Open-source Contributions

`care` github.com/LopezGroup-ICIQ/care Automated chemical reaction network generator for heterogeneous catalysis powered by ML estimators and integrated with kinetic solvers for multi-scale modeling

`GAME-Net` gitlab.com/iciq-tcc/nlopez-group/gnn_eads Graph neural network (GNN) for predicting the adsorption energy of molecules on transition metal catalysts

`GAME-Net` doi.org/10.19061/GAME-NET public web demo

`FG-dataset` zenodo.org/records/10410523 Dataset of molecules adsorbed on metal catalysts simulated at density functional theory (DFT) level

`pymkm` github.com/LopezGroup-ICIQ/pymkm Microkinetic modeling simulator for thermal and electrochemical heterogeneous catalysis

## Publications

06/02/2026 **Morandi, S.**, *et al.*, An end-to-end framework for reactivity in heterogeneous catalysis. *Nature Chemical Engineering*, **2** (2026)

08/09/2025 Lorenzutti, F., Seemakurthi, R., Johnson, E., **Morandi, S.**, *et al.*, Microenvironment effects in electrochemical $CO_2$ reduction from first-principles multiscale modelling. *Nature Catalysis*, **8**, 905-918 (2025)

01/05/2023 Pablo-García, S., **Morandi, S.** *et al.*, Fast Evaluation of the Adsorption Energy of Organic Molecules on Metals via Graph Neural Networks. *Nature Computational Science* **3**, 433-442 (2023)

07/11/2023 Sabadell-Rendón, A., Kázmierczac, K., **Morandi, S.** *et al.*, Automated Multiscale Simulation Environment. *Digital Discovery*, **2**, 1721-1732 (2023)

11/01/2022 Pablo-García, S., Sabadell-Rendón, A., Saadun, A. J., **Morandi, S.** *et al.*, Generalizing Performance Equations in Heterogeneous Catalysis from Hybrid Data and Statistical Learning. *ACS Catalysis*, **12**, 1581-1594 (2022)

30/03/2021 **Morandi, S.** *et al.*, A roadmap for in silico development and evaluation of industrial NMPC applications: A practical case study. *Computers & Chemical Engineering* **150**, 107278 (2021)

## Outreach

23 – 27/09/2024 TotalEnergies MathiasDays 2024, invited speaker, oral contribution "Automated Chemical Reaction Networks Generation in Heterogeneous Catalysis"

14 – 19/07/2024 $18^{\text{th}}$ International Congress on Catalysis (Lyon, France), contribution "Fast Evaluation of the Adsorption Energy of Organic Molecules on Metals via Graph Neural Networks"

6/06/2023 "Blau De Prussia" Spanish-Catalan podcast episode: "LabGPT: Una IA amb molt quimica" ("LabGPT: An AI with a lot of chemistry")

13 – 20/08/2022 BASF International Summer Course 2022, selected among the final pool of 50 participants

27 – 29/03/2021 NCCR Catalysis 2021, oral contribution "Advances in the Application of Machine Learning Techniques to Heterogeneous Catalysis"

## Awards & Scholarships

24/07/2024 Juan Bertrán i Rusca Outstanding Publication Award from the Spanish Royal Society of Chemistry

09/2017 – 12/2019 Full scholarship for high grade point average, years 2017-2019, Politecnico di Milano (1.5k EUR)

09/2024 – 12/2024 NCCR Catalysis Catalyzer Program, obtained 5.5K EUR funding for stay in industrial lab at IBM

## Other

09/2009 – 11/2018 **Volleyball player** *Volley Novara (Italy)*
Middle blocker for Volley Novara team. Joined the National Serie B2 league for two years and the regional Serie C for 6 years, achieving one promotion and reaching the Play-Off finals every year.

2025/ – present **Volleyball player** *CEVOL Torredembarra (Spain)*
Middle blocker in the first team competing in the primera nacional league.